COER UNIVERSITY

DBMS LAB ASSIGNMENT | LAB SHEET 4: SQL OPERATORS

Arithmetic Operators

Question 1

Question: Perform addition of two numbers using +.

Answer:

```
SELECT 25 + 15 AS AdditionResult;
```

Explanation: Adds two numeric values.

Question 2

Question: Perform subtraction using -.

Answer:

```
SELECT 50 - 18 AS SubtractionResult;
```

Explanation: Subtracts second operand from first.

Question 3

*Question: Perform multiplication using *.*

Answer:

```
SELECT 12 * 8 AS MultiplicationResult;
```

Explanation: Multiplies two numbers.

Question 4

Question: Perform division using /.

Answer:

```
SELECT 100 / 4 AS DivisionResult;
```

Explanation: Divides numerator by denominator.

Question 5

Question: Calculate the total amount as Price × Quantity.

Answer:

```
SELECT ProductName, Price, Quantity, (Price * Quantity) AS TotalAmount FROM Product;
```

Explanation: Multiplies Price and Quantity columns for each row.

Logical Operators

Question 6

Question: Demonstrate AND with two conditions.

Answer:

```
SELECT * FROM Student WHERE City = 'Delhi' AND Fees > 65000;
```

Explanation: Filters rows where BOTH conditions are TRUE.

Question 7

Question: Demonstrate OR with two conditions.

Answer:

```
SELECT * FROM Student WHERE City = 'Delhi' OR City = 'Roorkee';
```

Explanation: Filters rows where EITHER condition is TRUE.

Question 8

Question: Demonstrate NOT with a condition.

Answer:

```
SELECT * FROM Student WHERE NOT (City = 'Delhi');
```

Explanation: NOT negates the truth value of the expression.

Question 9

Question: Find students belonging to a specific course AND city.

Answer:

```
SELECT * FROM Student WHERE Course = 'B.Tech CSE' AND City = 'Roorkee';
```

Explanation: Applies AND operator across Course and City columns.

Question 10

Question: Find employees belonging to either of two departments.

Answer:

```
SELECT * FROM Employee WHERE Department = 'IT' OR Department = 'HR';
```

Explanation: Selects employees in IT or HR departments.

Comparison Operators

Question 11

Question: Demonstrate the = operator.

Answer:

```
SELECT * FROM Student WHERE City = 'Delhi';
```

Explanation: Checks exact string equality.

Question 12

Question: Demonstrate the > operator.

Answer:

```
SELECT * FROM Employee WHERE Salary > 55000.00;
```

Explanation: Filters employees earning strictly more than 55,000.

Question 13

Question: Demonstrate the < operator.

Answer:

```
SELECT * FROM Product WHERE Price < 2000.00;
```

Explanation: Filters products priced below 2,000.

Question 14

Question: Demonstrate >= and <=.

Answer:

```
SELECT * FROM Employee WHERE Salary >= 50000.00 AND Salary <= 65000.00;
```

Explanation: Tests inclusive lower and upper salary bounds.

Question 15

Question: Demonstrate the <>/!= operator.

Answer:

```
SELECT * FROM Student WHERE City != 'Delhi';
```

Explanation: Filters rows where City is not equal to 'Delhi'.

Special Operators

Question 16

Question: Demonstrate BETWEEN for salary range.

Answer:

```
SELECT * FROM Employee WHERE Salary BETWEEN 50000.00 AND 65000.00;
```

Explanation: Selects values within an inclusive range.

Question 17

Question: Demonstrate IN for selecting multiple cities.

Answer:

```
SELECT * FROM Student WHERE City IN ('Delhi', 'Roorkee', 'Dehradun');
```

Explanation: IN tests if a column value matches any item in a discrete list.

Question 18

Question: Demonstrate NOT IN.

Answer:

```
SELECT * FROM Student WHERE City NOT IN ('Delhi', 'Meerut');
```

Explanation: Excludes students from cities in the list.

Question 19

Question: Demonstrate LIKE for names beginning with a particular letter.

Answer:

```
SELECT * FROM Student WHERE Name LIKE 'A%';
```

Explanation: LIKE 'A%' matches string values beginning with 'A'.

Question 20

Question: Demonstrate IS NULL and IS NOT NULL.

Answer:

```
SELECT * FROM Student WHERE City IS NULL;
SELECT * FROM Student WHERE City IS NOT NULL;
```

Explanation: Tests for missing (NULL) or present values.

Set Operations & Integrated Queries

Question 21

Question: Demonstrate UNION using two compatible queries.

Answer:

```
SELECT Name, City FROM Student UNION SELECT Name, City FROM Employee;
```

Explanation: UNION combines distinct results from two queries into a single result set.

Question 23

Question: Demonstrate INTERSECT where supported by the DBMS.

Answer:

```
SELECT City FROM Student INTERSECT SELECT City FROM Employee;
```

Explanation: Returns rows that appear in both result sets.

Question 24

Question: Demonstrate EXCEPT/MINUS where supported.

Answer:

```
SELECT City FROM Student EXCEPT SELECT City FROM Employee;
```

Explanation: Returns distinct rows from the first query that do not appear in the second.

Question 25

Question: Perform a problem using Arithmetic + Logical + Comparison + Special operators.

Answer:

```
SELECT EmpID, Name, Salary, (Salary * 12) AS AnnualSalary
FROM Employee
WHERE (Salary * 12) > 600000
  AND Department IN ('IT', 'Finance')
  AND City = 'Delhi'
  AND Salary IS NOT NULL;
```

Explanation: Combines arithmetic (*), comparison (>), logical (AND), and special (IN, IS NOT NULL) operators.